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PROJECT DERMACHECK

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PROJECT DERMACHECK

Table of Contents

1. DermaCheck's Goals

1.1. Non-AI Goals

The non-AI system aims to provide a UI that:

- helps the user to create a personal account,
- shows on the screen the image taken, and
- uses a database to store one skin lesion image.

Additionally, it aims to provide informative details on any pathological case to help patients become informed about their condition (if the results are positive), as well as medical information about any pathological skin cancer case, including the dermatologist's contact details for further examination.

1.2. AI-related Goals

The AI system aims to provide image classification based on a pre-trained Visual Transformer. This is required because there is no other non-AI system that provides classification with such high accuracy as the provided classification system.

2. Functional Requirements

2.1. User Registration, Authentication, and Login

- Req001 - The system shall allow users to register using an email and password
- Req002 - While registering, the user is notified to use a valid email address.
- Req003 - While registering, the user is notified to use a valid password with several special characters.
- Req004 - When the user registers, the system shall display the registration screen.
- Req005 - When the user authenticates, the system shall display the authentication screen.
- Req006 - When the user logs in, the system shall display the login screen.
- Req007 - When the user is logged in, the system shall display the home screen.
- Req008 - The system shall allow users to log in using their registered email and password.
- Req009 - The system shall hash and store user passwords securely before saving them in a database
- Req010 - The system shall generate a JWT (JSON Web Token) upon successful login.
- Req011 - The system shall verify user identity using the JWT token before giving access to restricted endpoints.
- Req012 - The system shall allow users to reset their password if they forget it.

2.2. Image Upload and Processing

- Req013 - While the user uploads images, the system shall display the upload screen.
- Req014 - The system shall allow users to upload images containing skin lesions
- Req015 - The system shall prevent users from storing images that do not contain skin lesions.
- Req016 - The system shall evaluate the quality of the image before processing.

- Req017 - The system shall apply any preprocessing technique to the images (noise reduction, resizing, rotation, normalization).
- Req018 - When the user uploads an image on the system, the system shall limit uploaded images to a maximum size of 1 Gigabyte.
- Req019 - The system shall restrict the uploaded image format to JPEG, PNG, and JPG when the user uploads an image on the system.
- Req020 - If the user successfully uploads an image on the system, the system shall generate a unique ID for each uploaded image and shall store its metadata in a database.

2.3. Skin Cancer Classification

- Req021 - When the user waits for the classification results, the system shall display the upload screen, which shows a message that the image is being processed.
- Req022 - The system shall classify images with skin lesions into predefined categories/classes.
- Req023 - The system shall use a pre-trained Visual Transformer AI model to classify images.
- Req024 - The system shall provide a classification accuracy score for each image
- Req025 - If the system cannot classify an image, the user is notified to re-upload another image.

2.4. Display of Results

- Req026 - The system shall allow users to see the results of the classification
- Req027 - The system shall provide the classification accuracy score for each image in the results screen.
- Req028 - The system shall provide additional information about the classification results (probability scores)
- Req029 - The system shall store all past results in a database linked to the user's account
- Req030 - If the system cannot display the results due to performance issues, the system notifies the user to wait for 10 seconds.
- Req031 - If the system cannot display the results due to connection issues, the system notifies the user to establish a connection to the internet and restart the system.

2.5. Additional Features

- Req032 - The system shall allow users to find the nearest dermatologist to request a second opinion.
- Req033 - The system shall automatically delete images after a specified retention period of 30 days.

3. Non-Functional Requirements

3.1. Performance

- NfReq001 - While the device's camera is open, the system shall complete the classification of a patient’s image within 3 seconds.
- Nfreq002 - While the system is in operation, the system shall support at least 10000 users without a decrease in performance.
- NfReq003 - If the system cannot perform fast enough, the system shall notify the user that performance issues occur.

3.2. Security & Privacy

- NfReq004 - The system shall require MFA (multi-factor authentication) for the users
- NfReq005 - The system shall apply encryption based on Advanced Encryption Standards (AES) for storing images and patient data
- NfReq006 - The system shall enforce HTTPS for all communication
- NfReq007 - The system shall limit repeated failed login attempts to prevent brute-force attacks
- NfReq008 - The system shall use JWT authentication to verify user access
- NfReq009 - The system shall block IP addresses dealing with DDoS attacks.

3.3. Usability

- NfReq010 - The system shall provide a user-friendly interface accessible via mobile devices, tablets, and desktops.
- NfReq011 - The system shall allow users to download their test results in PDF format.

3.4. Reliability

- NfReq012 - According to international standards, the system shall have an uptime of 99.9% of the operational time.
- NfReq013 - The system shall automatically restart in case of a system crash.

4. AI-related requirements (use their IDs)

4.1. Regulatory & Compliance

- NfReq014 - The system shall store and process the images and comply with the following regulatory authorities:
  - GDPR (EU)
  - HIPAA (USA)
  - PIPL (China)

4.2. Ethics & Explainability

- NfReq015 - The system shall ensure all AI-related predictions are explainable to meet AI ethics guidelines.

4.3. Interoperability

- NfReq016 - The system shall provide integration of API services with 3rd party software and devices.

4.4. Safety

- NfReq017 - The system shall provide fail-safe mechanisms to prevent misdiagnoses without human validation.
- NfReq018 - The system shall conduct diligent testing to minimize false positives and false negatives.

4.5. Fairness

- NfReq019 - The system shall not be biased toward any demographic group in any of the classification outcomes
- NfReq020 - The system shall ensure fairness by providing models trained on diverse datasets representing different skin tones, ethnicities, and demographic groups.

4.6. Trust

- NfReq021 - The system shall maintain a transparent and ethical AI framework to gain user trust.
- NfReq022 - The system shall provide mechanisms for the users to report incorrect classifications and receive human-reviewed analyses.
- NfReq023 - The system shall engage with the medical community for the validation and acceptance of AI-generated classifications.

4.7. Transparency

- NfReq024 - The system shall have an interpretable decision-making process of the AI model used, understandable for medical professionals and patients.
- NfReq025 - The system shall disclose all the data sources, training methodologies, and model limitations to the users.
- NfReq026 - The system shall provide clear explanations of the AI classifications, including the reasoning of the predictions.

4.8. Organizational Culture

- NfReq027 - The system’s development and maintenance team shall endorse ethical AI principles and best practices provided by healthcare technology
- NfReq028 - The system’s organization shall promote a continuous learning culture, improvement in AI fairness, accuracy, and reliability, as well as due diligence.
- NfReq029 - The system’s organization shall encourage collaboration between AI researchers, dermatologists, and regulatory bodies to enhance the system’s effectiveness.

5. Use Case Descriptions

Use case 1: User Registration		
ID	UC1	
Description	The users register using an email, password	
Actors	Users	
Stakeholders:	Users, Developers, Managers	
Pre-Conditions	The system shows a registration screen	
Success end condition:	The user successfully registered to the system by using email and password as credentials	
Failure end condition:	The user is not allowed to register with an email and a password	
Main Success Scenario		**Linked UCs**
1	The user opens the DermaCheck App and views the registration screen	
2	The user registers by using an email address as a username and a password, by clicking on the register button	
3	The system stores the information provided to the database and sends a confirmation email to the user	
Alternative Scenarios		
2.A1		
2.A2		
Exception Scenario		
2. E1	The user fails to register (existing credentials or invalid email address domain)	
2. E3	The user tries again to register with a new email address and password	SUC1
2. E3		

Use Case 2: User Login			
ID		UC2	
Description		User logins to the system with credentials	
Actors		System	
Stakeholders:		Users, Developers, Managers	
Pre-Conditions		The system shows the login screen	
Success end condition:		The user logs in to the system with credentials	
Failure end condition:		The user cannot login to the system with credentials	
Main Success Scenario			**Linked UCs**
1	The user sees the login screen, and logs in by entering the credentials and clicking on the login button		
2	The system grants access to the user		
3	The user sees the home screen		
Alternative Scenarios			
2. A1	The user does not provide the credentials		
2. A2	The system denies access to the user and prompts the user to reset password		SUC1
2. A3	The user can reset the password on the login screen, by clicking on the button reset password		SUC1
Exception Scenario			
1. E1			
1. E2			
1. E3			

Use Case 3: Load Image			
ID		UC3	
Description		If the user successfully uploads an image on the system, the system initiates pre-processing	
Actors		Users	
Stakeholders:		Users, Developers, Managers	
Pre-Conditions		The user has acquired a skin lesion photo	
Success end condition:		The user loads an image successfully, and the system initiates pre-processing	
Failure end condition:		The user cannot load an image; the system cannot initiate pre-processing	
Main Success Scenario			**Linked UCs**
1	The user clicks the button "load image" on the home screen		
2	The user selects an image from a folder on a local system		
3	The user confirms that the selected image will be loaded and approves processing according to local regulatory bodies		
4	The system receives the image		SUC3
5	The system creates a unique image ID, stores it in the database, and after confirms if the image is corrupted and only the image file		
Alternative Scenarios			
1. A1			
2. A2			
Exception Scenario			
4. E1	The user fails to load an image on the system		
4. E2	The system prompts the user to reupload the image (because it is corrupted)		
4. E3	The user uploads an image again		

Use Case 4: View Classification Results		
ID	UC4	
Description	The user views the classification results on the screen	
Actors	Users	
Stakeholders:	Users, Developers, Managers	
Pre-Conditions	The system shows the classification screen	
Success end condition:	The user views the classification results on the screen and their details	
Failure end condition:	The user cannot view the classification results on the screen	
Main Success Scenario		**Linked UCs**
1	The user clicks on the button "Show results"	
2	The system sends the loaded image to the Visual Transformer Model to classify images into predefined classes/categories	
3	The system receives the classification results from the model	SUC4
4	The system shows the classification results on the screen together with categories, explanations and ethics information	
5	The user views the classification results and probability scores on the screen	
Alternative Scenarios		
1. A1	The user clicks the "Print Results" button on the screen	
1. A2	The system renders a PDF copy of the results and stores locally to the user's device	
Exception Scenario		
2. E1	The user does not view the classification results on the screen	
2. E2	The system requests the user to load a new image by showing a message on the screen	
2. E3	The user loads a new image by clicking on the "load image" button	UC3

Use Case 5: View Dermatologists List		
ID	UC5	
Description	The user searches for and views the nearest dermatologist to seek a second opinion	
Actors	User	
Stakeholders:	Users, Developers, Clinicians	
Pre-Conditions	The user requested granted location access to the system	
Success end condition:	The user views a list/map of nearby dermatologists on the screen	
Failure end condition:	No dermatologists nearby could be found, or location access is denied	
<u>Main Success Scenario</u>		**Linked UCs**
1	The user clicks on the button “Find Near Me” on the screen	
2	The system requires location access from the user with a notification message on the screen	
3	The user grants location access by clicking on the button "Grant Location Access" (on browser or system settings)	
4	The system creates a list containing nearby dermatologists and their contact details based on the geolocation	SUC5
5	The system shows a list of dermatologists in the vicinity of the user's location	
6	The user selects one list element to get the corresponding contact information	
7	The system shows on the screen the selected dermatologist's details	
<u>Alternative Scenarios</u>		
1. A1	The user denies location access	
2. A2	The system notifies the user cannot provide any information without location access (geolocation error)	
<u>Exception Scenario</u>		
1. E1	The user does not see any list/map on the screen (due to not granted geolocation access)	SUC5
2. E2	The system recreates the list/map of the dermatologists on the screen (after granting geolocation access)	SUC5
2. E3	The user clicks again the button "Find Near Me" or exits the app	SUC5

Supporting Use Case 1: Reset Password		
ID	SUC1	
Description	The user requests to reset their password	
Actors	User	
Stakeholders:	Users, Developers, Managers	
Pre-Conditions	The system provides a reset link on the screen	
Success end condition:	The user clicks on the reset link and updates the password	
Failure end condition:	The system cannot reset the password	

Main Success Scenario		**Linked UCs**
1	The user clicks the reset password link, "Forgot Password" on the login screen	UC2
2	The user enters their registered email address	UC2
3	The system verifies that the email exists in the database	UC2
4	The system sends a password reset link to the user's email address	UC2
5	The user clicks the link found in the received email and is directed to a reset password screen	UC2
6	The system shows the reset password screen	UC2
7	The user enters a new password and submits it	UC2
8	The system confirms the password reset	UC2
Alternative Scenarios		
3. A1	The system notifies the user, "email address not found."	
3. A2	The user tries to enter the correct credentials	
Exception Scenario		
4. E1	The system fails to send the reset link	
4. E2	The user retries or contacts support	

Supporting Use Case 2: Fail to Login		
ID	SUC2	
Description	The system manages a failed login attempt and provides feedback or recovery options to the user	
Actors	User	
Stakeholders:	Users, Developers, Managers	
Pre-Conditions	The system shows the login screen	
Success end condition:	The user submits the correct credentials successfully	
Failure end condition:	The user submits the wrong credentials	
Main Success Scenario		**Linked UCs**
1	The system detects incorrect or missing login credentials	
2	The system displays an error message to the user: "Invalid email or password."	
3	The system displays options: "Try again" or "Reset password."	
4	The user selects "Try again."	
5	The user enters the correct credentials	
6	The system grants access to the home screen	
Alternative Scenarios		
1. A1	The system shows a link named "Register again" on the screen	
1.A2	The user registers with new credentials	
4. A1	The user selects "Reset Password."	SUC1
4. A2	The system shows the "Reset Password" screen with 2 fields "Type new Password" and "Confirm Password"	SUC1
4. A3	The user fills the 2 fields "Type new Password" and "Confirm Password"	SUC1
4. A4	The system confirms change of password	SUC1
4. A5	The user enters the correct credentials	SUC1
4. A6	The system grants access to the home screen	SUC1
Exception Scenario		
3. E1	The system cannot connect to the database to verify credentials	
3. E2	The system retries the connection to the database until successful	

Supporting Use Case 3: Check Image File Integrity		
ID	SUC3	
Description	The system receives the loaded image and stores it in the database with appropriate metadata	
Actors	User	
Stakeholders:	Users, Developers, Managers	
Pre-Conditions:	The user has selected and confirmed an image to load	
Success end condition:	The image and its metadata are stored in the database	
Failure end condition:	The image cannot be saved, or data is lost during storage	
Main Success Scenario		**Linked UCs**
1	The user loads an image on the system	UC3
2	The system receives the image file from the front-end	UC3
3	The system checks image integrity (file not corrupted)	UC3
4	The image and its metadata are stored in the database	UC3
5	System confirms storage was successful and returns the image ID	
Alternative Scenarios		
Exception Scenario		
2. E1	The system found that the image file is incomplete or corrupted	UC3
2. E2	The system rejects the image and returns an error	UC3
2. E3	The user gets notified that the system is image file is corrupted	UC3

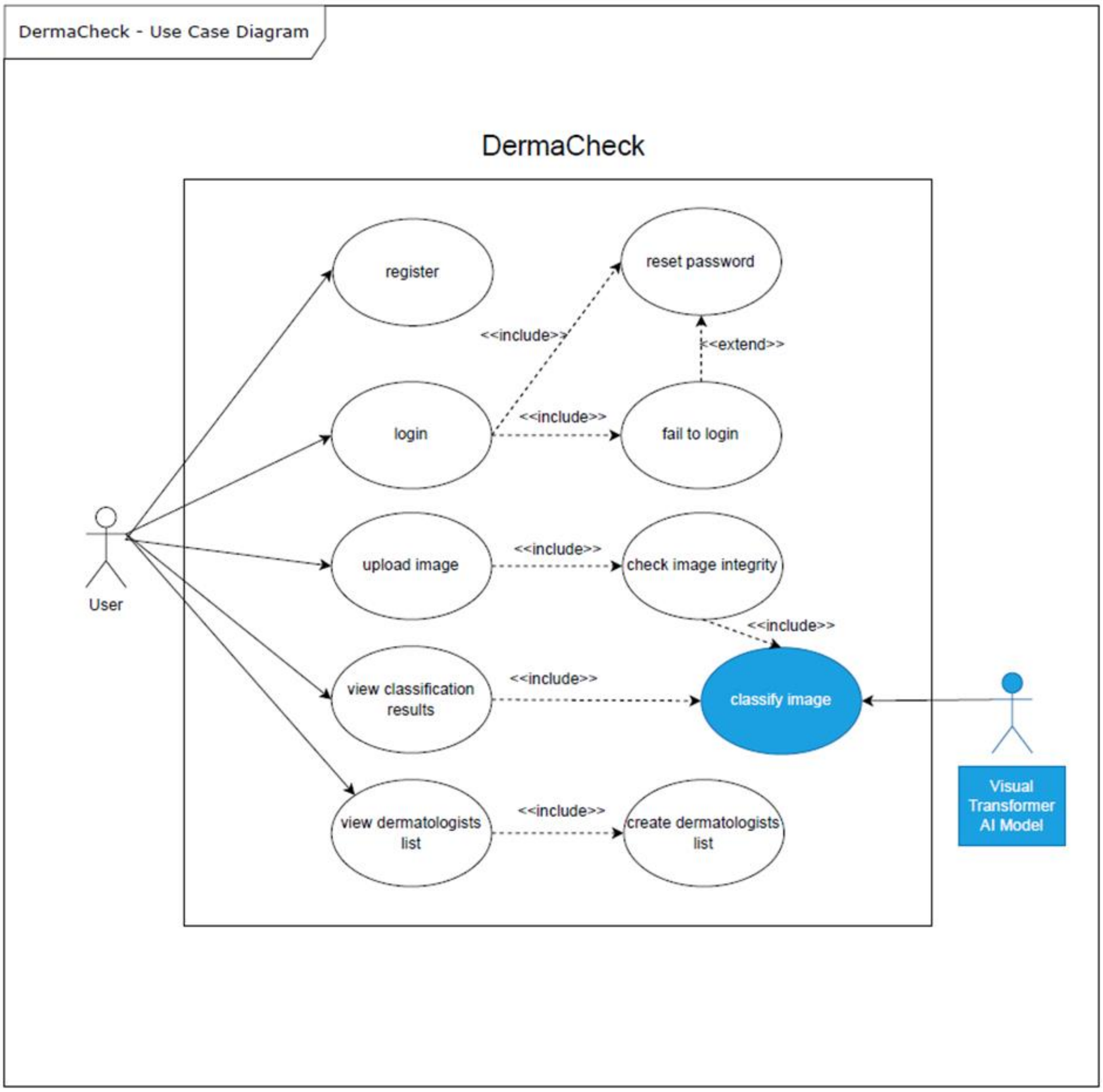
Supporting Use Case 4: Classify Image			
ID		SUC 4	
Description		The Visual Transformer classifies the loaded image	
Actors		Visual Transformer AI Model	
Stakeholders:		Users, Developers, Managers	
Pre-Conditions		The system loaded an image on the model	
Success end condition:		The model classifies successfully the loaded image	
Failure end condition:		The model failed to classify the loaded image and shows no results on the screen	
Main Success Scenario			**Linked UCs**
1	The system loads an image into the model		
2	The model classifies successfully the loaded image		
3	The system collects the results and shows them on the results screen		UC4
Alternative Scenarios			
Exception Scenario			
2. E1	The model failed to classify the loaded image and shows no results on the screen		UC3
2. E2	The system prompts the user that the model failed to classify the image by showing a notification on the screen		UC3
2. E3	The user loads another image on the system		UC3
2. E4	The system loads the image into the model		UC3

Supporting Use Case 5: Create List of Dermatologists		
ID	SUC5	
Description	The system retrieves nearby dermatologist locations based on the user's geolocation or entered area.	
Actors	User	
Stakeholders:	Users, Developers, Managers	
Pre-Conditions	The user has selected a dermatologist from the list shown in UC5.	
Success end condition:	The user submits the second opinion request.	
Failure end condition:	The system fails to send the request or form is incomplete	
<u>Main Success Scenario</u>		**Linked UCs**
1	The user clicks on the button "Show near me"	UC5
2	The system shows a list of links of nearby dermatologists,	UC5
3	The user selects one element of the list of dermatologists, and by clicking on each link	UC5
4	The system shows contact details and a map with the locations of the selected dermatologists on a browser	UC5
<u>Alternative Scenarios</u>		
1. A1	The user chooses an alternative element of the list of dermatologists	UC5
2. A2	The system shows the contact details of the selected alternative dermatologist	UC5
<u>Exception Scenario</u>		
2. E1	The user does not view any list of dermatologists on the screen and clicks again the button "Show nearby ..."	UC5
2. E2	The system shows a list of nearby dermatologists and a map with their locations on a browser	UC5

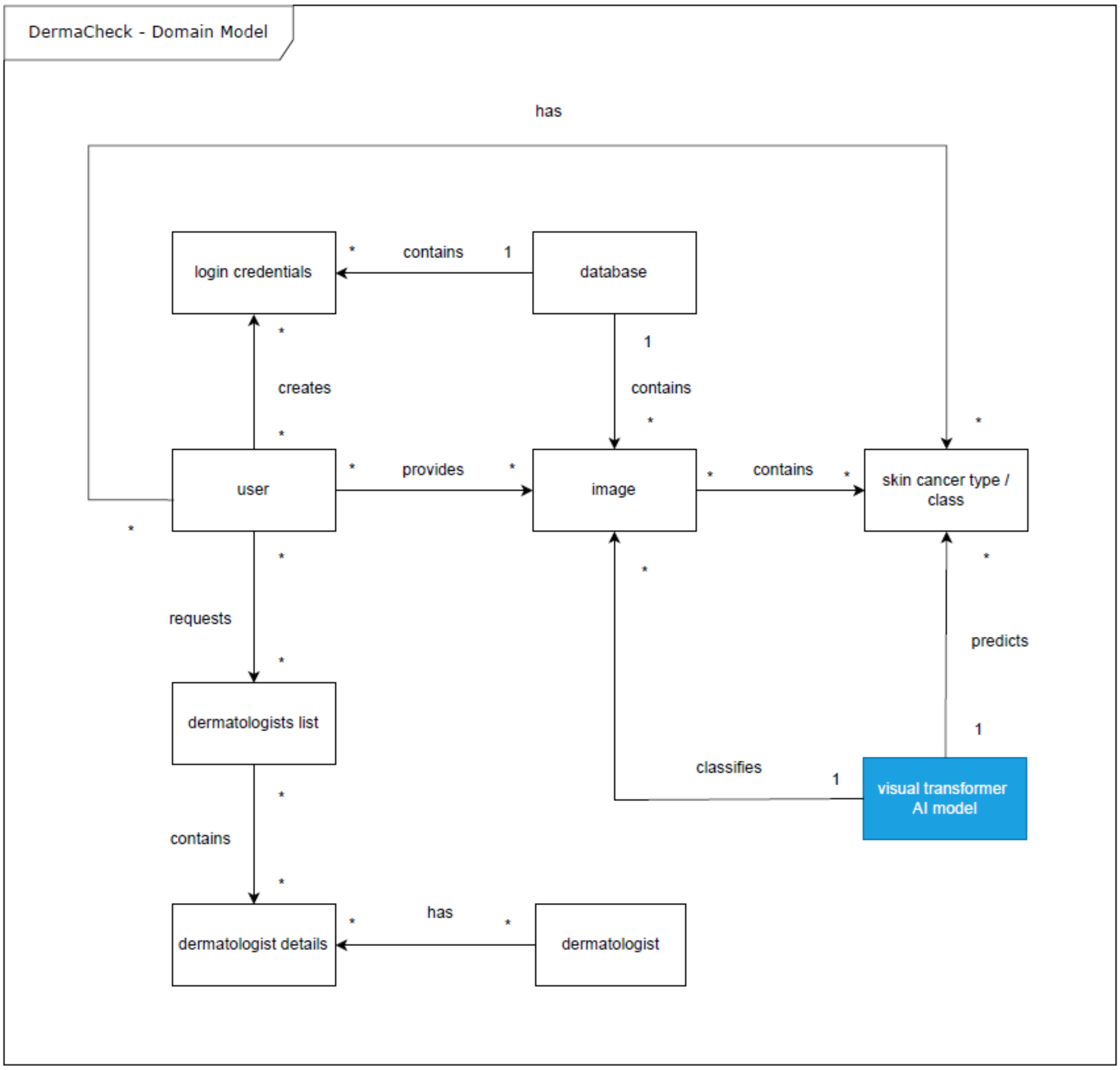
6. Traceability Matrix

Use cases	**UC1**	**UC2**	**UC3**	**UC4**	**UC5**	**SUC1**	**SUC2**	**SUC3**	**SUC4**	**SUC5**
Requirements										
Req001	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req004	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req006	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req008	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req012	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE
NewReq001	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
NewReq002	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
Req014	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req020	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req021	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req022	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
Req023	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
Req024	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
Req025	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req026	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req027	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req028	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req029	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Req032	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE
NfReq011	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
NfReq014	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
NfReq015	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
NfReq021	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
NfReq026	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
***NewReq001: The user fails to log in and the system provides links to register again or reset the password on the login screen**										
***NewReq002: The system checks if the image loaded is corrupt**										

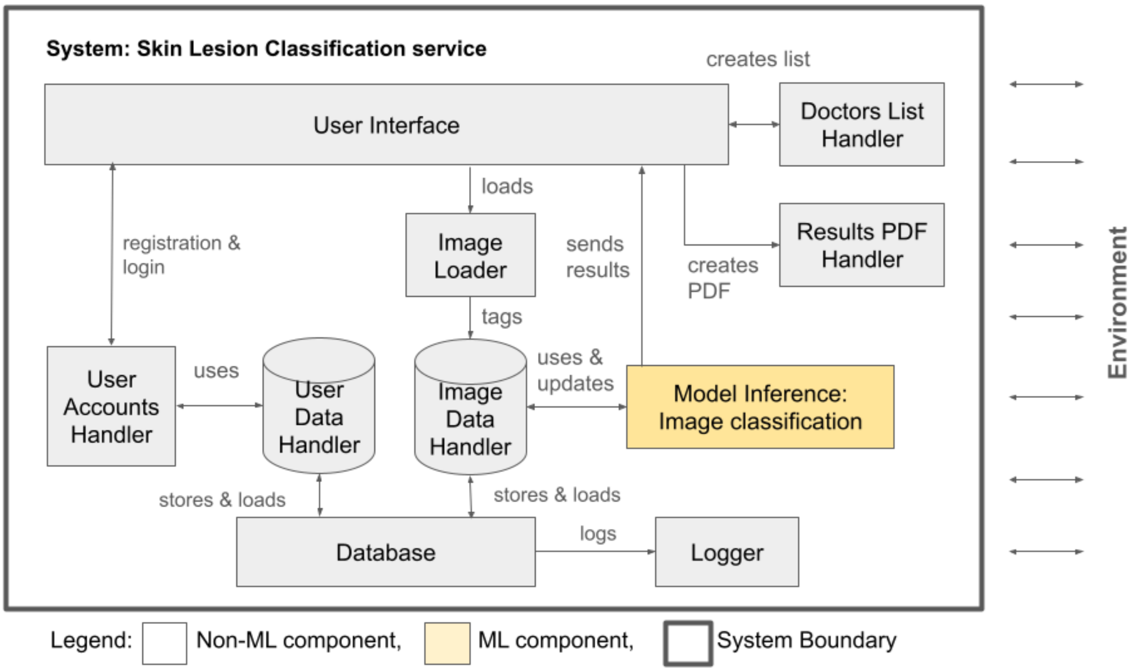
7. Use Case Diagram



8. Domain Model



9. Architecture Diagram



10. System Components Description

1. **User Interface:** It provides different screens depending on the content, which are assigned each time, like registration, login, home screen, and view results screen.
2. **User Accounts Handler:** Handles user accounts registration and secure access via login
3. **User Data Handler:** Stores personal information and links it to uploaded images. enables personalized history tracking.
4. **Database:** Stores all persistent data, including user profiles, images, and results, and allows logging of the activities to the logger.
5. **Image Loader:** handles image loading from the user interface to the database
6. **Image Data Handler:** provides a unique image ID to each image loaded and tags the images with metadata.
7. **Model Inference - Image classification:** classifies the images stored in the database and updates their metadata after the inference, sends results to the user interface
8. **Logger:** Records user actions and system events.
9. **Doctors List Handler:** creates lists of the nearest dermatologists
10. **Results PDF Handler:** creates a PDF with the results of the skin lesion classification

**We added a new requirement: (NewReq003: The system logs the actions and events)*

11. System Components Relation to Requirements

System Component	Req #	NfReq #
1. User Interface	004, 006, 026, 027, 028	
2. User Accounts Handler	001, 012, New001	
3. User Data Handler	008	
4. Database	029	
5. Image Loader	New002, 014	
6. Image Data Handler	020	014
7. Model Inference - Image Classification	021, 022, 023, 024, 025	015, 021, 026
8. Logger	NewReq003	
9. Doctors List Handler	032	
10. Results PDF Handler		011

Table 1: System components and their relation to the requirements

12. Design Questions

1. What is the first screen viewed by an unregistered user?
2. What are the character restrictions of the passwords used?
3. How many accounts can a user have on the system?
4. How many images can a user upload to the system?
5. How long is an image linked to a user’s account stored in the database?
6. How will we present the prediction results to the user?
7. How is the image data handler content-aware about skin lesions?
8. What should happen if the model fails to return a result?
9. Will the system support image upload via mobile devices?
10. What is safe accuracy for skin lesion prediction?
11. Is there a timeout period for the model's prediction?

12. What kind of information is logged to the logger?
13. How does the system find the nearest doctors' information?
14. What kind of information is printed in the PDF results?

13. Answers tot he Design Questions

1. The first screen a non-registered user views is the registration screen.
2. The system will require passwords to be at least 8 characters long and include at least one lowercase letter, one uppercase letter, and one number to ensure basic security.
3. Since a user account is based on the user's email address, for each unique email address, only one corresponding account is allowed.
4. A user can upload only one image per session (until the next image is loaded).
5. An image linked to a user's account is stored in the database only for 10 minutes, and its purpose is to be classified and used in the results screen and the PDF results. After this period
6. We'll display the predicted class, confidence percentage, and a link to the dermatologist's contact information.
7. The image data handler is aware of handling skin lesions based on the image classification. If the user loads a non-skin lesion image, they are notified that a new image should be uploaded.
8. The system will log the issue, notify the user with a friendly message, and allow a retry.
9. No, the system will allow users to upload existing images from their device storage. Direct camera capture will not be supported in this version.
10. A safe accuracy of skin lesion prediction could be >90%. Any lower, one might not be accurate enough to be considered for evaluation.
11. The timeout of the model would be 60 seconds.
12. The logger logs any action and system event based on database entries.
13. The system connects to Google Maps API and requests a list of dermatologists according to the user's location.
14. The PDF created by the system provides a copy of the loaded image, the model prediction, the probabilities per class, the regulatory and ethical aspects of the prediction, as well as information about the model's explainability.

14. Supplementary Information

14.1. Code Repository

https://github.com/dkastanis/DermaCheck

14.2. Presentation Video URL

https://drive.google.com/file/d/1H2u3XTjCwxfe3WJTKkYvpD4TKxtSJpee/view

14.3. AI Pre-trained Vision Transformer Model

The pre-trained Vision Transformer used in DermaCheck is offered by the HuggingFace user Anwarkh1 and the authors are grateful for this user's contribution and sharing on the HuggingFace platform:

Anwarkh1/Skin_Cancer-Image_Classification · Hugging Face

14.4. Image Discriminator Model

As part of the AI features implementation of DermaCheck, an extra feature was included in addition to the Vision Transformer, that is called “Image Discriminator”. This is an AI model that prevents (additionally to the Vision transformers capability) uploading non-skin lesion images on the DermaCheck app. It was trained to discriminate general images against skin-lesion ones, and its code and the model itself is included in the repository provided.

14.5. Gratitude to the Supervisors

We would like to thank Prof. Dr Alexander Egyed and Dr. Luciano Marchezan de Paula for the fruitful and interesting discussions as well as their valuable input during the JKU course of “Engineering AI-Intensive Systems”.